

***B.Tech. Degree IV Semester Examination April 2014*****CS 1403 COMPUTER ARCHITECTURE AND ORGANIZATION**  
(2012 Scheme)

Time : 3 Hours

Maximum Marks : 100

**PART A**  
(Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) What is the difference between big-endian and little-endian address assignment schemes?  
 (b) Differentiate between stacks and queues. Give one application of each data structures.  
 (c) What are guard bits? Discuss any two methods for truncating the guard bits.  
 (d) How does carry-save addition method improve speed of multiplication process?  
 (e) Why is a refreshing circuitry required in a DRAM organization in contrast to the SRAM design?  
 (f) Explain any one method for implementing the Least Recently Used (LRU) cache block replacement technique.  
 (g) What is cycle-stealing mode of DMA data transfer?  
 (h) Differentiate between memory mapped I/O data transfer and I/O mapped I/O data transfer.

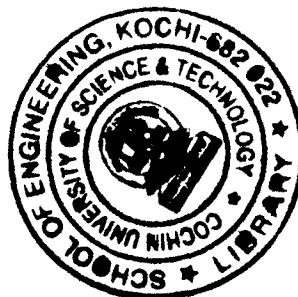
**PART B**

(4 × 15 = 60)

- II. (a) Describe the role of the following components of a digital computer with the help of a functional diagram of a computer. 10  
       (i) Memory Address Register  
       (ii) Program Counter Register  
       (iii) Memory Data Register  
 (b) How do condition codes and status flags affect the execution of program instructions? (5)
- OR**
- III. What are addressing modes? Describe any five addressing modes with suitable examples. (15)
- IV. (a) Write down the sequence of all control signals generated during the execution of the instruction, **Move (R1)+, LOC1**, where LOC1 is the address of a memory location. (10)  
 (b) What is a micro programmed control unit? (5)

**OR**

(P.T.O.)



- V. Explain the restoring and non-restoring division methods with a suitable circuit arrangement, where the divisor is the binary number **11** and the dividend is the binary number **1010**. (15)
- VI. (a) What is the importance of virtual memory concept? Explain the address translation mechanism used in the virtual memory implementation scheme. (10)  
(b) How do interleaved memory modules help in reducing the memory access time? (5)
- OR**
- VII. (a) What are the mapping functions associated with cache memories? Differentiate between associative and direct mapping techniques. (10)  
(b) Discuss the concept of Principle of Locality of reference (5)
- VIII. What is the need of a bus arbitration circuit in DMA controlled data transfer? How do centralized bus arbitration mechanisms differ from distributed arbitration mechanisms (15)
- OR**
- IX. (a) Describe the daisy-chaining arrangement of I/O devices (5)  
(b) What are vectored interrupts? (5)  
(c) Differentiate between synchronous and asynchronous schemes of data transfer. (5)

\*\*\*